

College Project Management & Review Portal (With Machine Learning)

Project Overview

This project is a web-based academic platform designed to manage the complete lifecycle of college projects. Students submit projects, tag preferred guides, receive reviews, and track progress digitally. Machine Learning is used to add intelligence and automation to the system.

Problem Statement

Traditional project handling in colleges is manual, fragmented, and inefficient. Students submit projects via emails or physical copies, guides lack progress visibility, and repeated project topics often occur. This leads to delays, miscommunication, and poor tracking.

Proposed Solution

A centralized portal where students submit projects and tag guides for review. Guides approve requests, review submissions, and provide structured feedback. Admins monitor progress and analytics without manual intervention.

User Roles

Student: Submit project, tag guide, upload reports, view feedback.

Guide: Approve/reject requests, review projects, assign marks.

Admin: Monitor users, deadlines, reports, and analytics.

Key Feature: Student Tags Guide

Students select a preferred guide during submission. The guide must approve the request before review begins. This ensures consent, workload balance, and realistic academic workflow.

Machine Learning Features

- Project Delay Prediction using Random Forest
- Duplicate Project Topic Detection using NLP (TF-IDF + Cosine Similarity)
- Student Performance Risk Analysis

Technology Stack

Frontend: React.js

Backend: Node.js / Flask

Database: MongoDB

ML: Scikit-learn, NLP techniques

Authentication: JWT

Benefits to College

- Organized project management
- Reduced delays and duplication
- Transparent evaluation process
- Early identification of weak projects

Conclusion

This project modernizes academic project handling using Full Stack development and Machine

Learning. It is scalable, practical, and directly applicable to real college environments.